

WEEK 09 | TUE 07·21 | ACT III REVIEW

The Hall, *Powered.*

You tuned the room for sound and negotiated with the sun. Now the sun is down and the switch is yours. This package is the evidence that your light levels were calculated, your luminaires were chosen as data, and the electricity behind them – load, distribution, and energy budget – was designed, not assumed.

ACT III ARC

Week 6 – Lighting kickoff: photometric quantities, the luminaire, the lumen method; first general-lighting pass. **Week 7** – Lighting design + controls in LightStanza; the full luminaire schedule; electrical distribution & load. **Week 8** – Panel schedule, circuiting, LPD compliance, and PV sizing. **Week 9** – Act III pin-up review; submit this package by Tue 07·21 at 11:59 PM.

DUE	WEIGHT
Tue 07·21	10%
11:59 PM · Canvas	20 pts raw
SUBMISSION	NAAB
PDF + Text	SC.4B
two parts, one assignment	+ SC.4E secondary

Act III closes with a two-part Canvas submission: a **PDF package** containing your lighting plan, lumen-method calculations, illuminance analysis, luminaire schedule, electrical load and panel work, energy figures, and pin-up slides; and a **pasted text entry** of your Decision Log entries from Weeks 6–8. Both parts are required. A missing Decision Log text entry will score zero on Criterion 5 regardless of the quality of the PDF.

The PDF is graded against a five-criterion rubric worth 20 points total. The central question for this act: *does the hall light as a set of layered systems that hit their targets – and does the electricity behind them hold together?* Each space must reach its IES illuminance target by calculation *and* simulation. Every luminaire must be a real product backed by photometric data. The load, panel schedule, LPD figure, and PV size must be consistent with the fixtures you actually specified.

LightStanza (or Revit's lighting analysis) is the required illuminance tool for this act. Results must appear as exported false-color maps or screenshots, not as verbal descriptions of what the software showed.

Export everything into a single PDF. Page order does not need to follow this list exactly, but all items must be present. Label each page or section clearly.

☐ **Lighting Plan / Reflected Ceiling Plan**

Plan (and RCP where it reads better) showing luminaire locations for the hall and prefunction, with the four lighting layers distinguished – house, egress/aisle, work/rehearsal, and architectural accent – and the control zone each is on. Include a fixture-type legend keyed to your schedule. North arrow required. Minimum 1/8" = 1' scale or equivalent.

☐ **Lumen-Method Calculations**

The hand calculation for your key spaces – at minimum the house and the prefunction, plus stage/work light if applicable. For each: the maintained target E (fc + lux) with the IES Lighting Handbook category cited; delivered lumens Φ , CU, and LLF with a one-line justification each; $N = (E \cdot A) / (\Phi \cdot CU \cdot LLF)$ solved, then N plugged back to report the achieved E ; and the installed watts ($N \times W$) for the space.

☐ **Illuminance Analysis – LightStanza / Revit**

False-color illuminance maps (footcandles or lux) for your primary spaces, exported from LightStanza or Revit. Annotate each map with the target level and the achieved average, and note whether the simulation agrees with your lumen-method hand calc within reason. If it does not, identify the most likely cause (CU assumption, reflectances, fixture spacing, surface losses).

☐ **Luminaire Schedule**

A schedule of every fixture type, each a real specified product backed by an IES file. For each type list: delivered (fixture) lumens Φ , input watts, CCT, CRI, distribution type, and mounting. Tag each type to the layer and task it serves (e.g., warm wide-diffuse house downlight; high-CRI accent on stage; narrow direct over aisles; asymmetric cove wash). Use delivered lumens, not lamp lumens.

☐ **Electrical Load Summary + Panel Schedule**

The connected lighting load summed by space and by circuit (W and VA / amps), grouped into logical circuits (by layer, zone, or scene). A panel schedule listing each circuit with its load, breaker size, and phase, plus the total connected load and a reasonable demand. Acknowledge other building loads (receptacles, mechanical) with allowances even if not fully detailed.

☐ **LPD Compliance Check**

Installed Lighting Power Density (W/ft²) for the hall and/or whole building, compared explicitly to the ASHRAE 90.1 / IECC allowance for the space type, with a clear pass/fail. If you are over the

allowance, state the design response (higher-efficacy fixtures, fewer fixtures, or a controls credit).



PV Sizing Calculation

Size a rooftop PV array to offset a stated fraction of the building's lighting (or total) load. Show the target offset, the array size in kW, the module count, and the approximate roof area required, with your assumptions noted (module wattage, local peak sun-hours). A back-of-envelope sizing is expected here, not a full production model.



Act III Pin-Up Slides

Your presentation from the Week 9 in-class review, exported as PDF pages. Graded on content, not slide design. A concise 6–8 slides covering: the layered lighting concept, illuminance results vs. targets, the luminaire schedule, the LPD/PV energy story, and the one cross-system trade-off you navigated (where lighting met an Act I acoustic surface or your Act II daylight strategy).

In the Canvas text entry box below the file upload, paste the full text of your Decision Log entries from Weeks 6–8 (your Act III lighting and electrical entries). Do not summarize or edit – paste as written. The Week 6 lighting entry (the first lumen-method pass) is the minimum required entry for this act; later entries should document how your illuminance results, LPD figure, or load work caused you to revise, confirm, or abandon a fixture or distribution choice.

FORMAT FOR THE TEXT PASTE

Copy each entry from your Google Doc exactly as written. Separate entries with a blank line. You do not need to reformat.

Entry 9 • Week 6

Decision: House lit with 14 warm 3000K downlights at 4,000 lm delivered...

System: Lighting

Space: Main Hall (house)

Reasoning: Target 12 fc between acts; first pass at 18 fixtures came in at 11 fc, so I raised Φ from 3,500 to 4,000 lm and cut to 14 to hold target and trim installed watts...

Trade-off: The accent cove that grazes my Act I reflector cloud shares a dimmer with the house layer – will need a separate scene...

Reconsider: If the LightStanza map shows dark aisles, add a dedicated egress layer rather than raising the house level...

Entry 11 • Week 8

Decision: Installed LPD came in at 0.78 W/ft² against a 0.90 allowance – pass...

...

Five criteria, four points each, on a five-level scale: *Exceeds* (4) · *Mastery* (3) · *Near* (2) · *Below* (1) · *No Evidence* (0). Scores map directly to the gradebook: 20 raw points = 10% of the course grade. Each criterion is evaluated independently. *Mastery* represents full competency at the graduate level for this stage of the project.

01 Lighting Design & *Illuminance*

SC.4B

SC.4E

4 pts

EXCEEDS 4	Lumen-method calculations are shown for the key spaces (house, prefunction, and stage/work light), each with a cited IES maintained target (fc + lux) and the achieved E reported. Results are verified against a LightStanza or Revit false-color illuminance map; any discrepancy between hand calc and simulation is explained. A reflected ceiling or lighting plan shows the four layers – house, egress, work, accent – each on its own control zone, with fixture locations and a legend. The plan reads as a designed system, not a uniform grid.
MASTERY 3	Lumen-method calcs are present for at least two spaces with targets cited and achieved E reported. An illuminance map is included. The lighting plan shows the layers and controls with minor gaps – one layer or control zone is unclear, or simulation and hand calc are not directly compared.
NEAR 2	A single calc or a generic illuminance map without comparison to a target. The plan shows fixtures but does not differentiate the layers or their controls. Lighting is treated as one level for the whole room.
BELOW 1	No lumen-method calculation, or no lighting plan. Fixtures are placed without a target illuminance or any analysis to support the layout.
NO EVIDENCE 0	Nothing submitted addressing lighting levels or layout.

02 Luminaire Selection & *Photometric Logic*

SC.4B

4 pts

EXCEEDS 4	A luminaire schedule lists real, specified fixtures backed by IES files. Each row gives delivered (fixture) lumens Φ , input watts, CCT, CRI, distribution type, and mounting. Every fixture is matched to its job: narrow/direct over aisles and apron; wide/diffuse warm light for the house and lobby; high-CRI (≥ 90) accent on the stage; asymmetric wash on architectural surfaces. Delivered lumens – not lamp lumens – are used in the calculations.
MASTERY 3	Schedule lists at least three real fixtures with most specifications present. Distribution, CCT, and CRI are mostly matched to task. Delivered lumens are used. Minor gaps – one fixture missing a spec, or one task/distribution mismatch unaddressed.
NEAR 2	Fixtures are named but the schedule is incomplete (no IES data, or CCT/CRI/distribution not stated). Matching of fixture to task is weak or asserted without reasoning.
BELOW 1	Generic fixtures with no specifications and no schedule. No photometric data of any kind.
NO EVIDENCE 0	No luminaire selection submitted.

03 Electrical Distribution & *Load*

SC.4E

4 pts

EXCEEDS 4	Connected lighting load is summed by space and by circuit (W and VA/A). Circuits are grouped by a logical principle – by layer, zone, or control scene – and the grouping is explained. A panel schedule lists each circuit with its load, breaker size, and phase, and reports the total connected load and a reasonable demand. Other building loads (receptacles, mechanical) are acknowledged with allowances even if not fully detailed.
MASTERY 3	A load summary and a panel schedule are present with minor gaps – breaker sizing is approximate, one zone is missing, or demand factors are not applied. Circuit grouping is reasonable.
NEAR 2	A partial load list or a circuiting diagram, but no organized panel schedule. Loads are not totaled, or grouping logic is absent.
BELOW 1	Loads are mentioned without calculation, or no distribution work beyond naming a panel.
NO EVIDENCE 0	No electrical distribution work submitted.

04 Energy Performance — *LPD & PV*

SC.4B

SC.4E

4 pts

EXCEEDS 4	Installed Lighting Power Density (W/ft ²) is calculated and compared explicitly to the ASHRAE 90.1 / IECC allowance for the space type, with a clear pass/fail. If over, a specific design response is proposed (higher-efficacy fixtures, fewer fixtures, controls credit). A PV array is sized to offset a stated fraction of the building's lighting (or total) load: the calculation shows the target offset, array size in kW, module count, and approximate roof area required, with assumptions (panel wattage, sun-hours) noted.
MASTERY 3	Both an LPD comparison to an allowance and a PV sizing calculation are present, with minor gaps in derivation or stated assumptions.
NEAR 2	One of the two (LPD <i>or</i> PV) is addressed but not both, or a value is stated without a calculation behind it.
BELOW 1	Energy is mentioned without any numbers – no LPD figure, no PV size.
NO EVIDENCE 0	Neither LPD nor PV is addressed.

05 Decision Log *Quality*

SC.4B

4 pts

EXCEEDS 4	Four or more entries covering Weeks 6–8. At least one entry references a specific number or result (e.g., “house came in at 11 fc against a 12 fc target with 18 downlights; raised Φ to 4,000 lm and cut to 14 fixtures to hold target and trim installed watts”). At least one entry names a cross-system trade-off: a lighting layer that reveals or fights an Act I acoustic surface, an electric layer coordinated with the Act II daylight-harvesting strategy, or an LPD constraint that changed a fixture choice. Reasoning is in the student’s own voice and shows that data drove the decision.
MASTERY 3	Three or more entries covering Weeks 6–8. At least one connects a decision to a calculation or result. At least one cross-system trade-off is acknowledged. Trade-off field completed on most entries.
NEAR 2	Two entries, or entries describe what was done without referencing analysis results. No illuminance, LPD, or load numbers appear. No acknowledgment of the link to acoustics or daylight.
BELOW 1	Zero or one entry, or entries are placeholder text. No connection between the log and the lighting/electrical work in the PDF package.
NO EVIDENCE 0	No Decision Log entries submitted for Act III.

This assignment addresses two Student Performance Criteria from the NAAB 2020 Conditions. Criteria 2 and 5 map primarily to SC.4B; Criteria 1, 3, and 4 draw on both environmental and building-service systems.

SC . 4 B

Environmental Systems

Principles of environmental systems design; application to problems involving daylighting, lighting systems, and acoustics; tools used for performance assessment; analysis across different geographic regions.

SC . 4 E

Building Service Systems

Principles of building service systems including electrical distribution and lighting; appropriate application of quantitative performance analysis; use of simulation technologies to assess environmental performance.

Use these as calibration guides, not guarantees. A student who scores Mastery on all five criteria earns 15/20 (B range), which is the expected outcome for a graduate student running the lumen method and a first electrical pass for the first time.

SCORE	GRADE EQUIV.	WHAT THE WORK SHOWS
18 – 20	A	Light levels are calculated <i>and</i> verified in simulation; the luminaire schedule is real and task-matched; the panel schedule, LPD figure, and PV size hang together with the fixtures specified. The Decision Log shows data driving design and at least one genuine cross-system trade-off – a lighting layer reconciled with an Act I surface or the Act II daylight strategy.
14 – 17	B	Solid technical work with a minor gap in one criterion. Lumen-method calcs and a luminaire schedule are present; an illuminance map is included; load and energy work are started and mostly consistent. Log is substantive but may lack a cross-system trade-off entry.
10 – 13	C	Calculations or schedules present but disconnected – illuminance without simulation verification, fixtures without specs, or LPD/PV stated without derivation. Lighting reads as a uniform level rather than layered systems. Log entries describe what was done rather than why the numbers changed the approach.
6 – 9	D	Major gaps in technical content – no lumen-method calc, no luminaire schedule, or no electrical/LPD work. Decision Log missing or one entry only. The PDF does not demonstrate that calculation drove any lighting or electrical decision.
≤ 5	F	Did not meaningfully engage with the lighting/electrical problem. No illuminance output, no schedule, required elements absent. Contact the instructor immediately if you are in this range before the deadline.